ELSEVIER

Tectonophysics 292 (1998) 85–100

TECTONOPHYSICS

New evidence of clockwise vertical axis rotations south of the Arica elbow (Argentine Puna)

Claudia B. Prezzi *, Juan F. Vilas

Laboratorio de Paleomagnetismo 'Daniel Valencio', Departamento de Ciencias Geológicas, Universidad de Buenos Aires, Ciudad Universitaria, Pabellón II, 1428 Buenos Aires, Argentina

Received 21 March 1997; accepted 27 January 1998

Abstract

The Central Andes show a pronounced bend at 18°S. This bend is called the Arica elbow. The existence of counterclockwise vertical axis rotations north of the Arica elbow and clockwise rotations to the south has been determined from palaeomagnetic data. Various authors have proposed different mechanisms to explain this, but whether rotations are local or regional and whether the Arica elbow is a primary or secondary feature of the Central Andes remains controversial. In order to contribute to the palaeomagnetic data available for the Argentine Puna, 102 oriented cores were obtained from Cenozoic rocks cropping out in the Bajo de las Siete Curvas area of the southern Argentine Puna. Different demagnetization techniques were applied. Pretectonic remanent magnetizations were isolated, and indicate the existence of a clockwise vertical axis rotation of approximately 7°–18°. Taking into account the presence of a major thrust fault to the west of the studied area, this rotation could be considered to be of local character, induced by a Miocene–Pliocene compression. © 1998 Elsevier Science B.V. All rights reserved.

Keywords: palaeomagnetism; Cenozoic; Puna; rotation; clockwise; local

1. Introduction

The Central Andes show a pronounced bend at 18°S. This bend is called Arica elbow.

The existence of counterclockwise vertical axis rotations north of the Arica elbow and clockwise rotations to the south, has been determined from palaeomagnetic data. These rotations have been found in Mesozoic and Cenozoic rocks cropping out in the Andean forearc and backarc (Butler et al., 1984; Isacks, 1988; Beck, 1988; MacFadden et al., 1990, 1993, 1995; Beck et al., 1994; Somoza, 1994; Singer et al., 1994; Butler et al., 1995; Somoza et al., 1996a,b; Randall et al., 1996; etc.). This spatial distribution of rotations is what Somoza et al. (1996a) called the Central Andes Rotation Pattern (CARP).

It remains controversial whether these rotations are local or regional, and whether the Arica elbow is a primary or secondary feature of the Central Andes. Furthermore, various models have been proposed by different authors (Carey, 1955; Beck, 1988; Isacks, 1988; Dewey and Lamb, 1992; Gubbels et al., 1993; Beck et al., 1994; Watts et al., 1995; Somoza et al., 1996a,b; Randall et al., 1996; etc.) in order to define the driving process or processes leading to the CARP, but a consensus has not been reached. These models predict clockwise vertical axis rotations for the Andean foreland relative to the South

* Corresponding author. Fax: +54 1 788 3439.

0040-1951/98/$19.00 © 1998 Elsevier Science B.V. All rights reserved.
PII S0040-1951(98)00058-4

America Craton, south of the Arica elbow. However, palaeomagnetic data (MacFadden et al., 1990, 1995; Roperch et al., 1993; Singer et al., 1994; Butler et al., 1995; Aubry et al., 1996; etc.) demonstrate that rotations involve not only the continental margin, but also the Puna and Altiplano.

One of the most important problems to deal with when trying to define the processes leading to the CARP is the scarcity of palaeomagnetic data. This problem is critical for the region of the Argentine Puna that extends approximately between 22° and 27°S (Somoza et al., 1996b). There is a complete lack of palaeomagnetic data for this region.

Here, the final results obtained from Cenozoic rocks cropping out in the southern Argentine Puna are presented. The sampling zone is close to Salar de Pocitos and extends between 24°30′ and 24°40′S and 67°00′ and 67°15′W (Fig. 1). In this region,

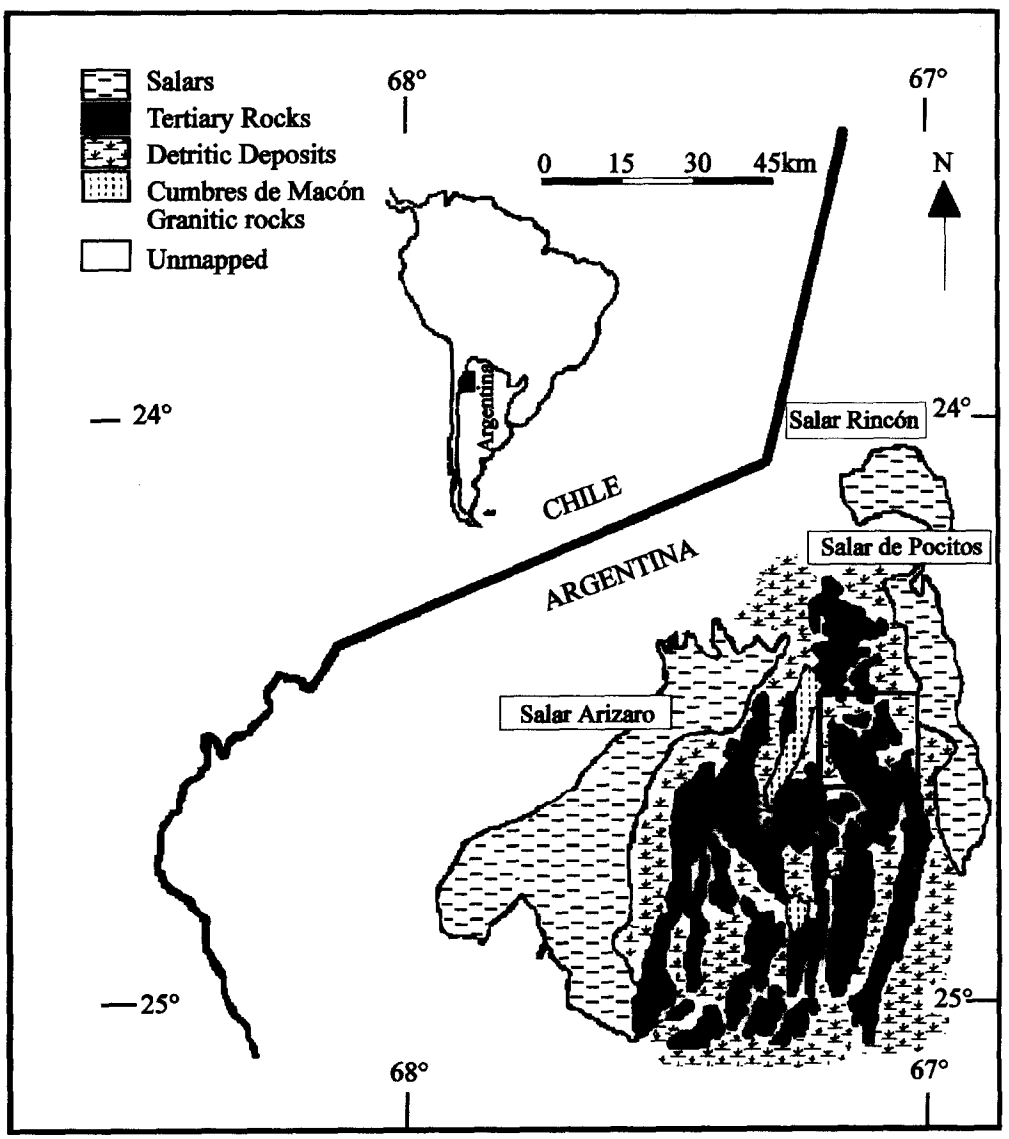


Fig. 1. Location map of the study area showing distribution of the Tertiary rocks sampled. Box west of Salar de Pocitos outlines area shown in Fig. 2.

102 oriented samples of Tertiary sedimentary rocks were collected for this palaeomagnetic study. The aim of this study is to determine the existence and sense of any vertical axis rotation in the area, and if possible, also its regional or local character. Some speculations are made about the possible origin of these vertical axis rotations.

2. Geologic setting

2.1. Stratigraphy

At present we do not know of any detailed geological study of this region of the Argentine Puna. The rocks sampled have not yet received formational names, nor have they been correlated with known formations. General descriptions of this sedimentary section are given in some papers (e.g. Alonso et al., 1991; Vandervoort, 1993; Vandervoort et al., 1995).

Three different informal units were identified in the sampling zone, based on lithological changes observed from base to top of the section. Their distribution and sampling locations are shown in Fig. 2.

Unit A: the lowest unit, consisting mainly of 10–20 cm thick strata of red mudstones (70%) with interlaminated gypsum. Fine (16%), medium (8%) and coarse (5%) sandstones are interlayered along the section. At the top of the section are pyroclastic sedimentary rocks (1%); a tuff yielded a hornblende $^{40}Ar/^{39}Ar$ age of 23.8 ± 0.4

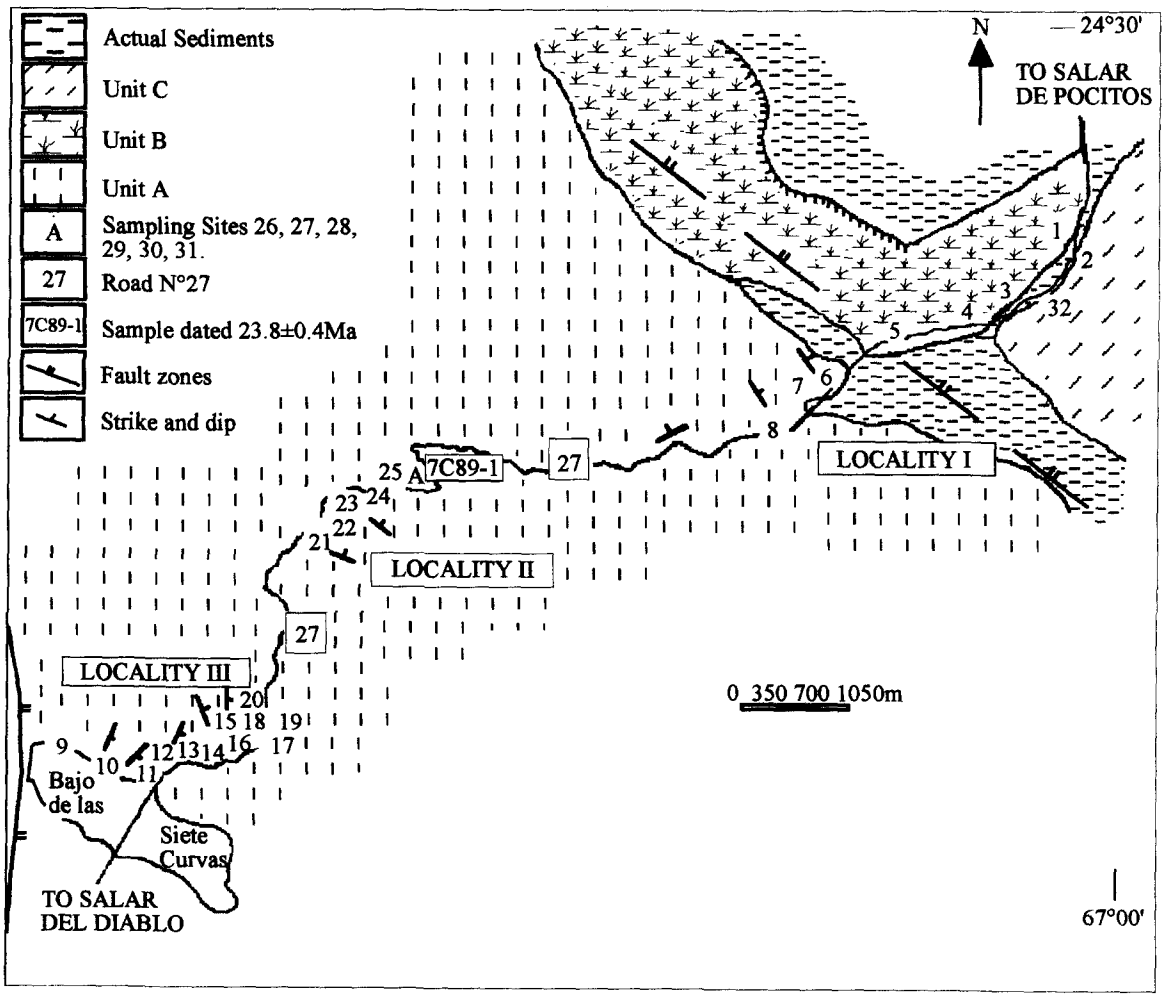


Fig. 2. Simplified geologic map of the sampling zone. Localities and sites sampled are shown.

Ma (Late Oligocene–Early Miocene) (Vandervoort, 1993) (sample 7C89-1; Fig. 2). No fossils were found. The depositional environment was arid continental, probably dominated by playa lakes.

Unit B: the middle unit, consisting mainly of 15–30 cm thick strata of pink, grey and pale-brown medium pyroclastic sandstones (70%). Red and grey fine sandstones (15%) and pink, pale-brown and grey mudstones (10%) are frequently interlayered; at the top of the section pale-brown conglomerates are observed (5%). A tuff near the top of the section yielded a zircon fission track age of 10.8 ± 2.0 Ma (Middle to Late Miocene) (Alonso et al., 1991). The depositional environment was continental arid to semiarid and this unit probably was deposited in a sandy braided fluvial system. The contact between units B and A is masked by a fault zone, but it seems to be conformable (Vandervoort, 1993).

Unit C: the highest unit, consisting mainly of coarse to very coarse conglomerates (70%) interlayered with medium conglomerates, coarse sandstones and sandstones (30%). The section shows many different colors (red, green etc.) which are due to the pyroclastic origin of most of the sediments. The contact with unit B is angular–unconformable.

2.2. Structure

Till now there are no detailed structural or kinematic studies of the area available. The most important features are two fault zones. One of them was observed at the west side of El Bajo de las Siete Curvas (to the west of the area shown in Fig. 2); its general strike is NNE–SSW. The block west of this fault zone, where pre-Tertiary sedimentary rocks crop out, was uplifted. This fault zone is recognized as the southern end of a high-angle thrust fault that was identified to the north of the study area by other authors (e.g. Méndez, 1974; Fernández Garrasino et al., 1984; Koukharsky, 1988) as the eastern boundary of the granitic rocks of Cumbres de Macón (Fig. 1). These rocks have a biotite K/Ar age of 670 ± 20 Ma (Koukharsky, 1988). These granitic rocks have been interpreted by Fernández Garrasino et al. (1984) as occupying a block which has been transported eastward along a thrust fault which shallows at depth. Koukharsky (1988) concluded that the Cumbres de Macón block underwent a tilt of 70° westward, and is overthrust on Tertiary sedimentary rocks. Koukharsky (1988) considered this fault to be the result of east–west compression, but also that it has an important strike-slip component, and suggested that this fault developed during the Principal Quechua Phase of deformation, near the Miocene–Pliocene boundary. The other fault zone was observed in the northeast quadrant of the sampling zone (Fig. 2). Its general strike is NW–SE. The south block, where unit A crops out, was uplifted.

3. Palaeomagnetic study

3.1. Field work

Three different localities were sampled and 102 oriented cores from 32 sites were collected using a portable gasoline-powered drill (water was used as a coolant for the drill bit). Each locality was defined because of attitudinal changes in the strata (Fig. 2).

Locality I: units A, B, and C cropping out. General attitude: strike 340°, dip 12° NE. The thickness of the sampled section at this locality was about 300 m (unit A 150 m, unit B 100 m, unit C 50 m). 30 samples were collected from 9 sites.

Locality II: unit A cropping out. The bedding is subhorizontal. The thickness of the sampled section at this locality was around 110 m. 40 samples were collected from 11 sites.

Locality III: unit A cropping out. General attitude: strike 20°, dip 13° SE. The thickness of the sampled section at this locality was around 360 m. 32 samples were collected from 12 sites.

The youngest outcrops of unit A are in Locality I, the oldest ones are in Locality III.

3.2. Treatment procedures

One to three standard specimens from each sample were cut in the laboratory. To measure the magnetizations three different magnetometers where used: a modified DIGICO spinner magnetometer, a Schonstedt SSM2 magnetometer and a 2G cryogenic magnetometer. Detailed alternating fields and thermal demagnetization techniques were applied. With the first technique, up to 26 demagnetization steps were performed, using the 2G cryogenic magnetometer's alternating field demagnetizer, and reach-

ing fields of 170 mT. With the second one, up to 20 steps were performed, reaching temperatures of 700°C in a Schonstedt TSD-1 furnace. The bulk magnetic susceptibility of the specimens was measured after each heating step in order to monitor possible magnetic mineral changes.

Experiments of IRM (isothermal remanent magnetization) acquisition were performed by applying increasing magnetic fields in successive steps, reaching fields of 2300 mT. For a more conclusive identification of magnetic minerals, thermal demagnetization of the IRM imparted along three orthogonal axes was carried out (Lowrie, 1990) applying fields of 120, 410 and 2400 mT, respectively. Polished sections from different specimens were observed using a reflecting microscope.

3.3. Obtained data

Locality III. Most specimens (50%) had very low unblocking temperatures (about 150–200°C) and coercivities (about 20–30 mT). These specimens showed random magnetization changes upon thermal or alternating field demagnetization. Some specimens (30%) showed very high unblocking temperatures and very high coercivities. The remanence intensity remained practically constant through demagnetization, although alternating fields were applied up to 70 mT and temperatures of 680°C were reached. Total demagnetization was not possible. The polarity of these specimens is reverse and the vectors representative of the remanent magnetizations are clustered in the southwest quadrant of the stereonet. The specimens showed a dramatic increase in their bulk magnetic susceptibility (about 200–500%) after heating steps in the range 570–680°C (Fig. 3). Few specimens (20%) showed high unblocking temperatures and high coercivities. Remanence intensity decreased through demagnetization, reaching 10–20% of the original value after applying temperatures of 500–570°C or magnetic fields of 35–40 mT. These specimens showed demagnetization paths going through normal and reverse polarity magnetizations. The normal polarity magnetizations correspond to an overprint probably related to deformation events. The specimens showed a dramatic increase in their bulk magnetic susceptibility (about 200–500%) after heating steps in the range 570–680°C (Fig. 4).

Locality II. Most specimens (40%) showed low unblocking temperatures and low coercivities. Remanence intensity decreased quickly through demagnetization, reaching 10–20% of the original value after applying temperatures of 400–500°C or magnetic fields of 35 mT. These specimens showed demagnetization paths going through normal and reverse polarity magnetizations. The normal polarity magnetizations correspond to an overprint probably related to deformation events. The bulk magnetic susceptibility remained almost constant during the successive steps of thermal demagnetization (Fig. 5). Remanence intensity of some specimens (30%) decreased slowly after each step of thermal or AF demagnetization, reaching 10–20% of the original value after applying temperatures of 680–700°C or magnetic fields of 170 mT. These specimens show normal polarity, and vectors representative of the remanent magnetizations are clustered in the northeast quadrant of the stereonet. The bulk magnetic susceptibility remained almost constant during the successive steps of thermal demagnetization. Magnetizations are single-component (Fig. 6). Some specimens (30%) had very low unblocking temperatures (150–200°C) and coercivities (20–30 mT). These specimens showed random behavior upon thermal or alternating field demagnetization.

Locality I. Specimens from units B and C showed anomalous and random behavior and were rejected. Most specimens (65%) from unit A showed low unblocking temperatures and low coercivities. Remanence intensity decreased quickly through demagnetization, reaching 10–20% of the original value after applying temperatures of 300–400°C or magnetic fields of 35 mT. These specimens showed demagnetization paths going through normal and reverse polarity magnetizations. The normal polarity magnetizations correspond to an overprint probably related to deformation events. The bulk magnetic susceptibility remained almost constant during the successive steps of thermal demagnetization. Remanence intensity of some specimens (35%) decreased slowly after each step of thermal or AF demagnetization, reaching 10–20% of the original value after applying temperatures of 680–700°C or magnetic fields of 170 mT. These specimens have normal polarity, and vectors representative of the remanent magnetizations are clustered in the northeast quadrant of

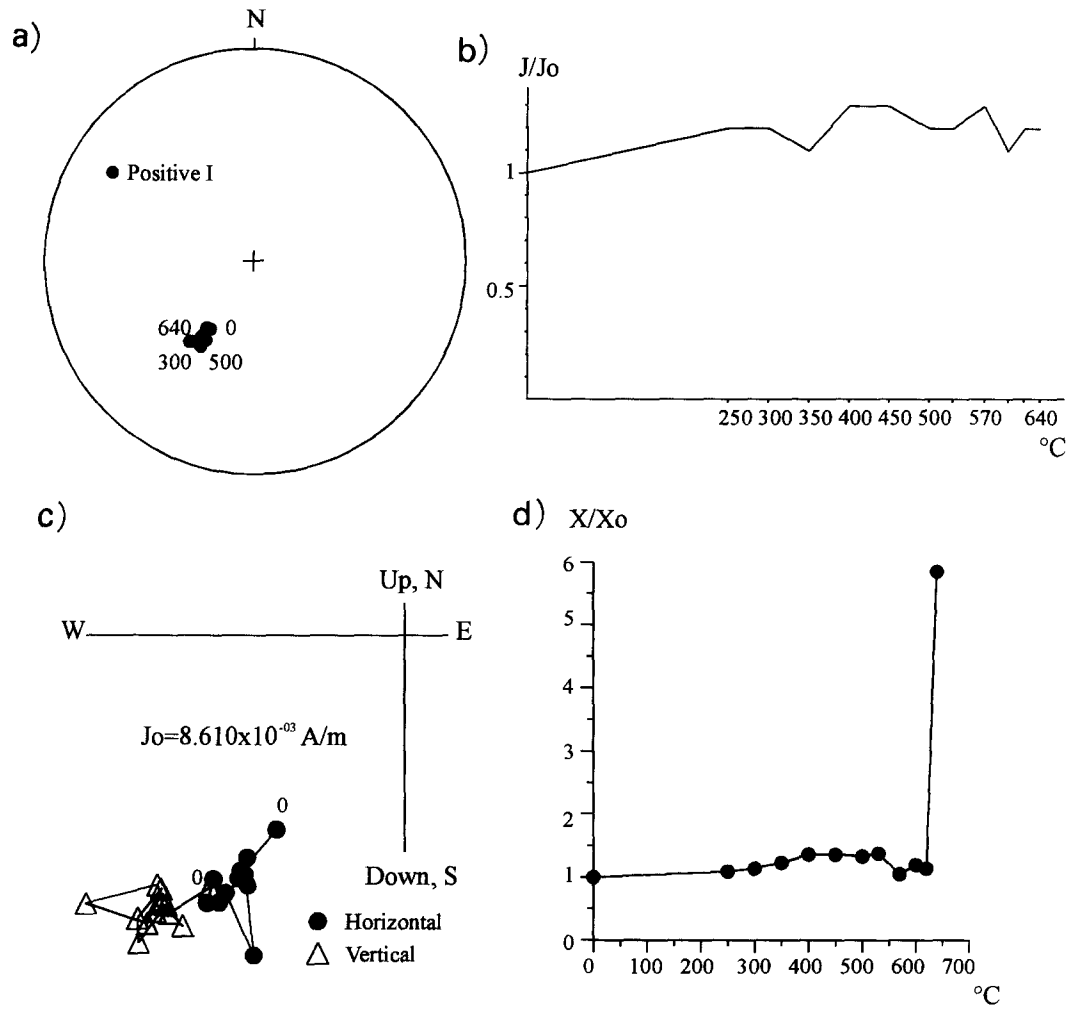


Fig. 3. Results obtained after thermal demagnetization of specimen A02911.SPC from Locality III. (a) Stereographic representation of each remanent magnetization vector. (b) Normalized demagnetization curve. (c) Zijderveld diagram. (d) Variation of bulk magnetic susceptibility (X) through thermal demagnetization (normalized with bulk magnetic susceptibility at room temperature (Xo)).

the stereonet. The bulk magnetic susceptibility remained almost constant during the successive steps of thermal demagnetization. Magnetizations are single-component.

The experiments of IRM acquisition showed that the main carrier of the remanence is hematite. This is indicated by the lack of saturation. The steep slope of the curve in the first steps suggests the additional presence of some (titano)magnetite (Fig. 7). The thermal demagnetization curves of composite IRM show an abrupt drop in the soft, medium, and hard coercivity fractions around 550°C, but even after 700°C total demagnetization is not achieved. There is another abrupt drop in the soft, medium and hard coercivity fractions around 350°C. These changes in slope of the thermal demagnetization curves indicate the presence of magnetite, hematite and perhaps titanomagnetite or titanohematite, as carriers of the remanence (Fig. 8).

Because of the increase in bulk magnetic susceptibility observed in specimens of Locality III, experiments of IRM acquisition were performed on two different specimens obtained from the same oriented core, one of which had undergone thermal demagnetization up to 620–720°C. This procedure was applied for samples from Locality III and for sam-

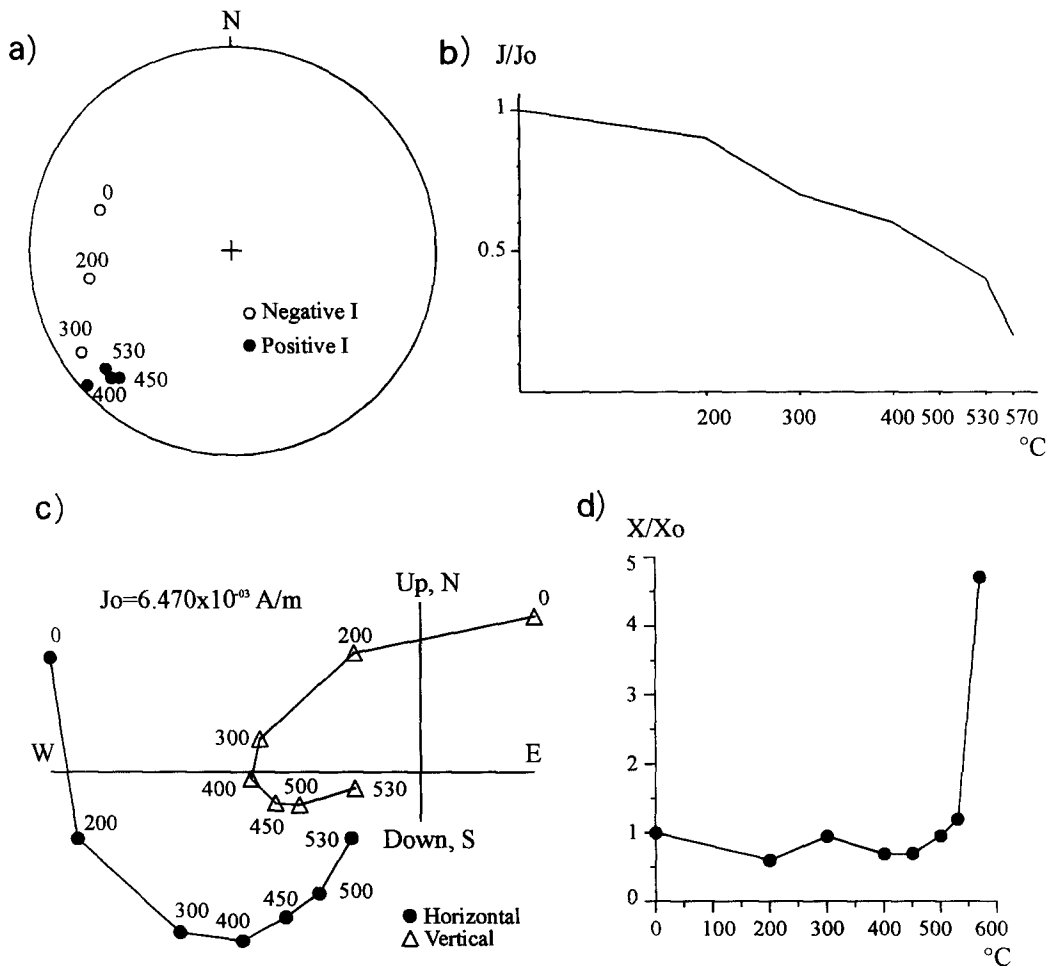


Fig. 4. Results obtained after thermal demagnetization of specimen A04311.SPC from Locality III. (a) Stereographic representation of each remanent magnetization vector. (b) Normalized demagnetization curve. (c) Zijderveld diagram. (d) Variation of bulk magnetic susceptibility (X) through thermal demagnetization (normalized with bulk magnetic susceptibility at room temperature (Xo)).

ples from Locality II. In the case of the samples from Locality III, specimens which showed bulk magnetic susceptibility increases when subjected to thermal demagnetization, showed higher saturation intensities than specimens not subjected to thermal demagnetization (Fig. 9a and b), suggesting that new magnetic minerals formed during the successive heating steps. IRM acquisition diagrams for specimens of Locality II showed the same saturation intensity for specimens subjected to thermal demagnetization as for those that did not undergo heating (Fig. 9c and d), indicating no formation of new magnetic minerals. In order to analyze in more detail the coercivity spectra derived from the IRM curves, the incremental isothermal remanent moments (ΔJ mA m^{-1}), in 100 mT intervals (ΔH, applied magnetic field) were calculated for both heated and unheated specimens, and plotted as histograms (Dunlop, 1972). Dunlop (1972) studied the coercivity spectra of red beds and divided them into soft ($0 < H < 100$ mT, maximum remanent coercivity), intermediate ($100 < H < 300$ mT) and hard ($300 < H < 1800$ mT) components. He considered that these components indicate the relative contributions to IRM of magnetite, specularite and pigment, respectively. When comparing the two histograms obtained for each sample, the most important changes were observed in the magnetite ($0 < H < 100$ mT) fractions for specimens com-

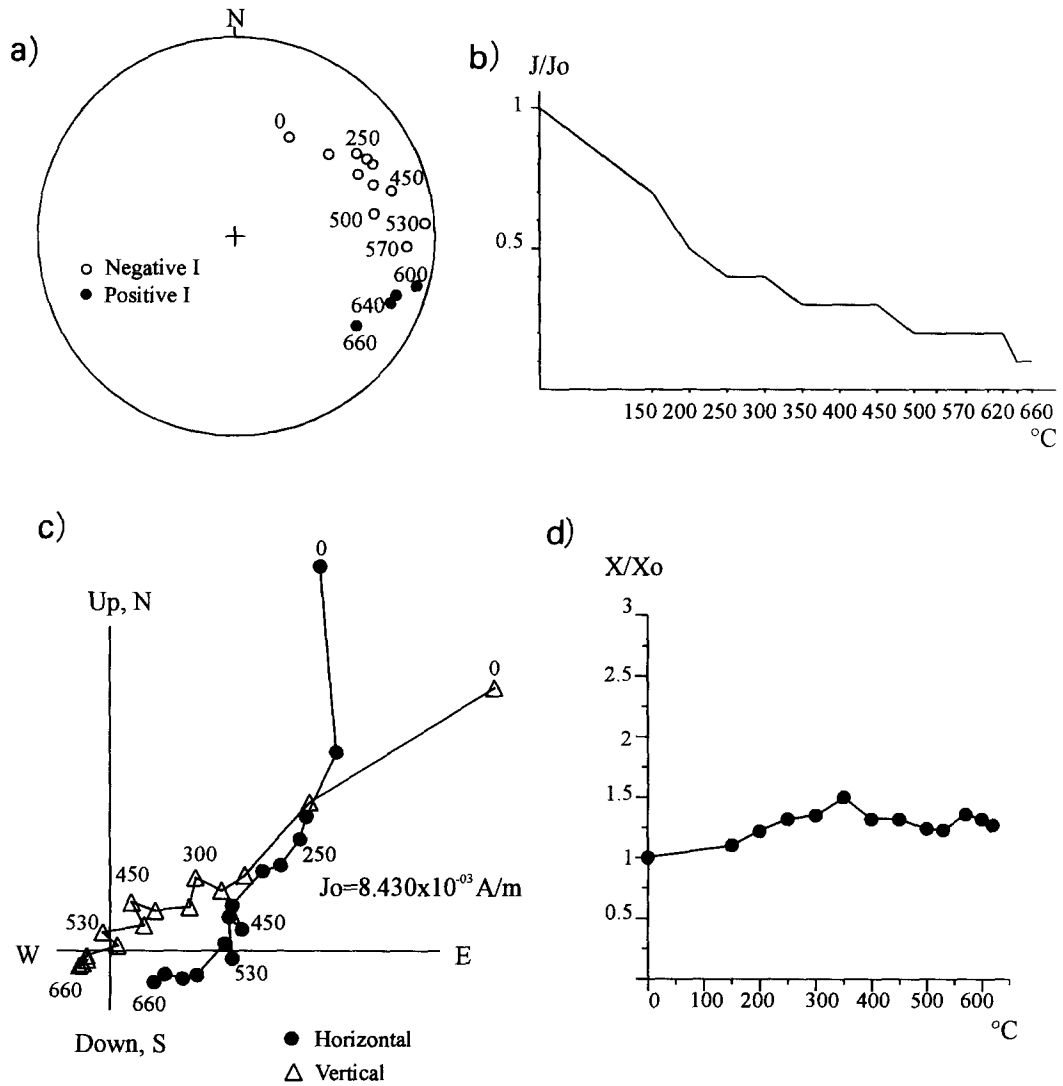


Fig. 5. Results obtained after thermal demagnetization of specimen A08811.SPC from Locality II. (a) Stereographic representation of each remanent magnetization vector. (b) Normalized demagnetization curve. (c) Zijderveld diagram. (d) Variation of bulk magnetic susceptibility (X) through thermal demagnetization (normalized with bulk magnetic susceptibility at room temperature (Xo)).

ing from Locality III samples (Fig. 10a and b). For these specimens it was determined that heating to 620°C and more resulted in magnetite production (Turner, 1980). The magnetite generation on heating of red beds has often been reported in the literature (Stephenson, 1967; Schwarz, 1969, 1970; etc.) and is probably due to the breakdown of iron-silicates (Dunlop, 1972). These iron-silicates perhaps could occur in the matrix clay fraction. When comparing the histograms obtained for specimens not subjected to thermal demagnetization from Locality II with those from Locality III, it can be observed that the amount of magnetite present is much greater in the Locality II samples than in those from Locality III (Fig. 10a and c), while the pigment fraction ($1000 < H < 3000$ mT) is almost equal for all specimens. These results would indicate that the main magnetic carrier in the Locality II samples is magnetite, while in those from Locality III it is hematite. Taking into account these assumptions, the different

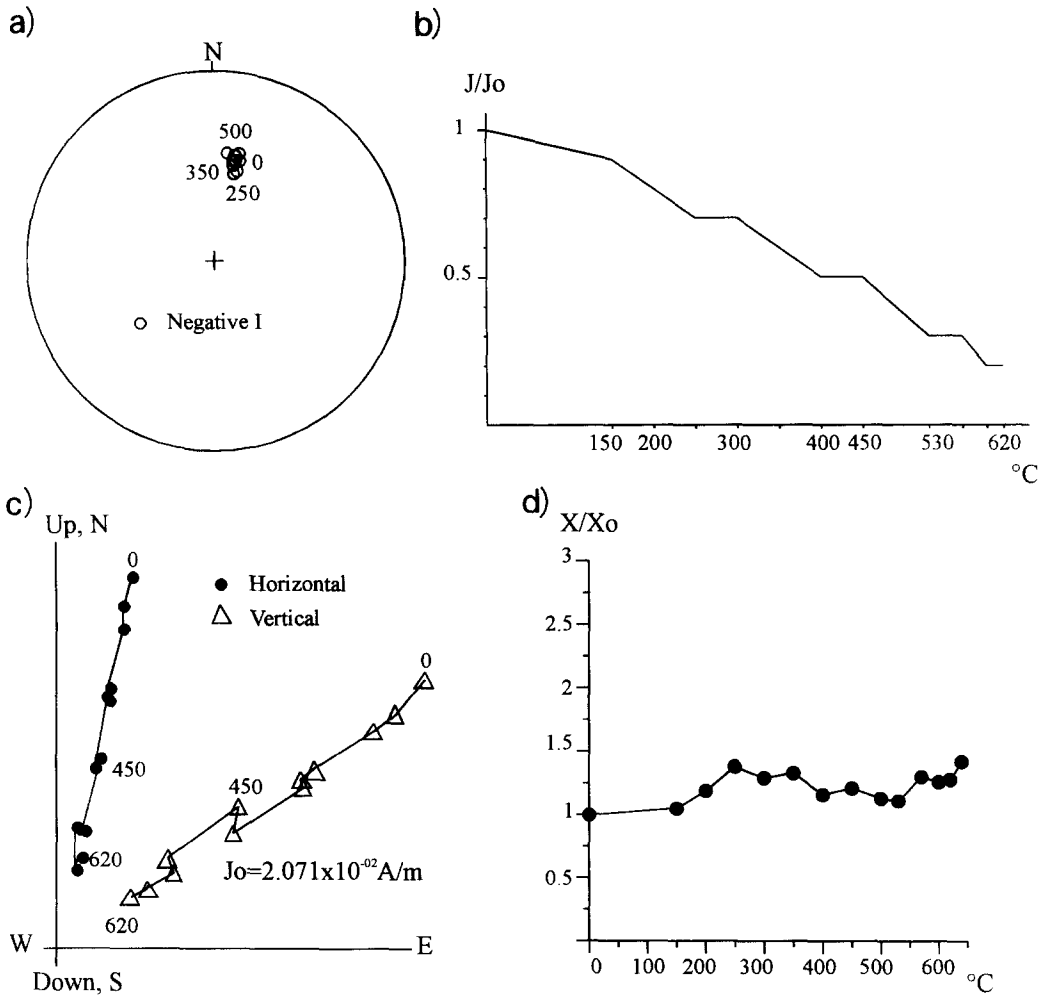


Fig. 6. Results obtained after thermal demagnetization of specimen A06711.SPC from Locality II. (a) Stereographic representation of each remanent magnetization vector. (b) Normalized demagnetization curve. (c) Zijderveld diagram. (d) Variation of bulk magnetic susceptibility (X) through thermal demagnetization (normalized with bulk magnetic susceptibility at room temperature (Xo)).

behaviors shown by the samples from Locality II and those from Locality III when subjected to different demagnetization techniques, could be explained in terms of a distinct participation of different magnetic minerals in the rock's composition.

From observations of polished sections of some specimens it was determined that (titano)magnetite, hematite, ilmenite, limonite, pyrite and gold are present. Scarce (titano)magnetite crystals of nearly 100 μm in diameter and more numerous ones of 10 to 30 μm in diameter were observed. These crystals are of detrital origin. It was not possible to determine the Ti content of (titano)magnetite. Hematite was identified as an alteration (limonitization) product. Hematite exsolution textures were observed in detrital ilmenite crystals of nearly 30 μm in diameter. Limonite was recognized as a weathering product. Very scarce pyrite of nearly 10–20 μm in diameter and gold of nearly 5 μm in diameter were observed in each polished section. The identification of these minerals coincides with the results obtained from the IRM acquisition experiments, from which the presence of (titano)magnetite, hematite and perhaps titanomagnetite or titanohematite was inferred. The sizes of these minerals, determined through observations of the polished sections, indicate the

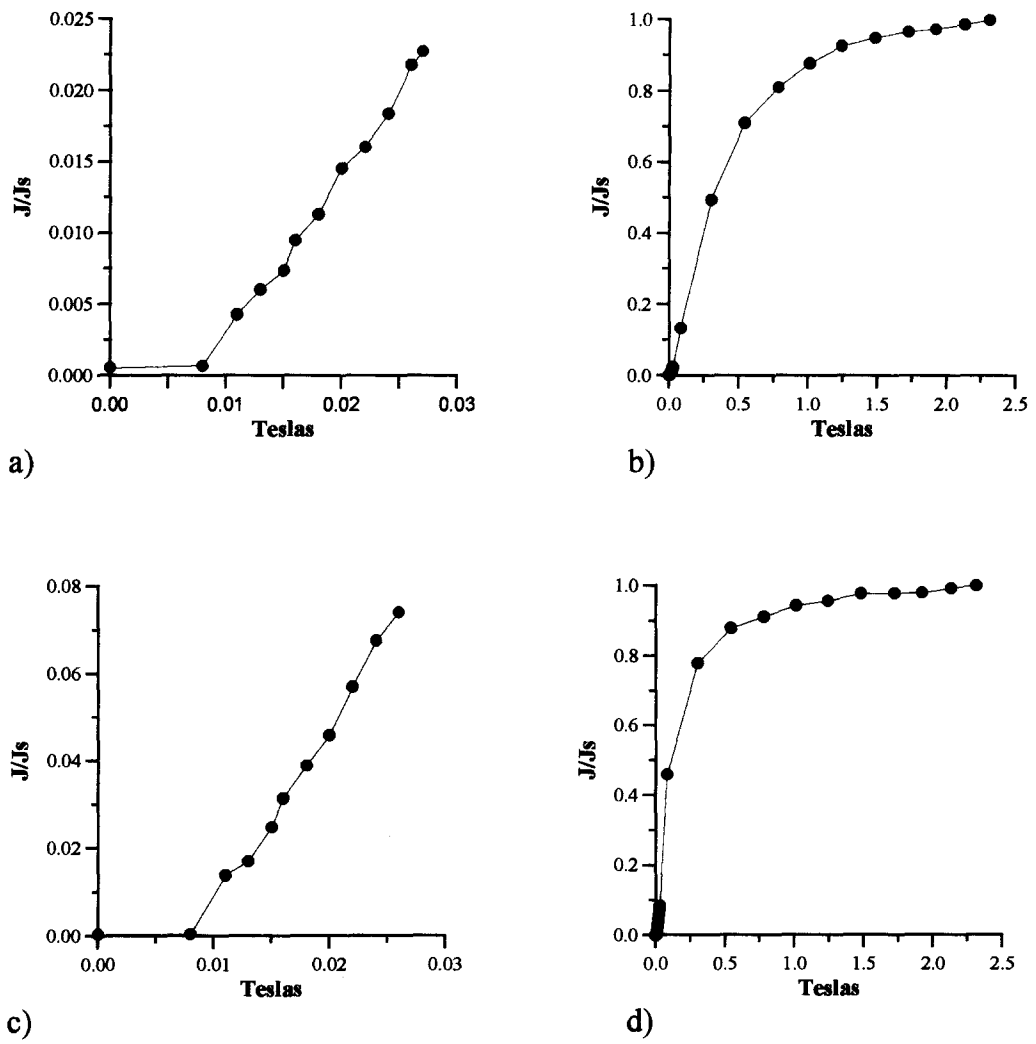


Fig. 7. IRM acquisition curves for specimen A05312.SPC from Locality III (a, b) and A01911.SPC from Locality I (c, d). (a and c) Results in low fields. (b and d) Results in high fields.

presence of multi-domain and pseudo-single-domain magnetite and probably single- or pseudo-single-domain titanohematite (Dankers, 1978, among others). Pseudo-single-domain magnetite could explain the abrupt drop observed in the thermal demagnetization curves of composite IRM around 550°C, while the abrupt drop observed around 350°C could be due to the presence of titanohematite or titanomagnetite. Perhaps multi-domain magnetite could be the principal magnetic carrier in the samples that showed low unblocking temperatures and coercivities and random behavior on demagnetization.

3.4. Data analysis

Specimens with very low unblocking temperatures were discarded because of their random behavior. Specimens were analyzed using the computer program MAG88 (Oviedo, 1989). Magnetic directions were plotted on vector end point diagrams (Zijderveld, 1967). The corresponding components of natural remanent magnetizations (NRMs) were determined using principal component analysis (Kirschvink, 1980) when the linear trends of the vector end points were identified. For demagnetization

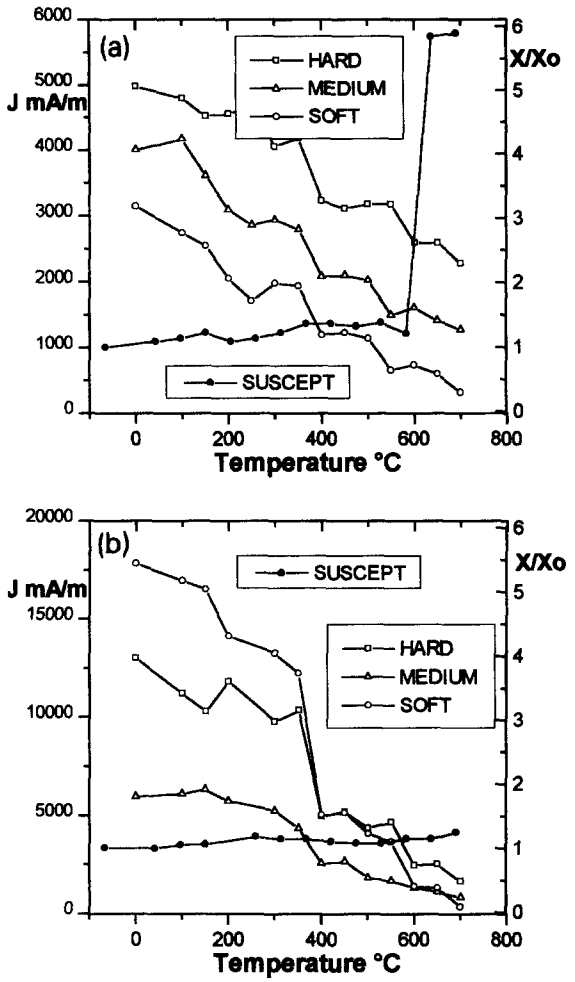


Fig. 8. (a) Thermal demagnetization curve of composite IRM for specimen A02913.SPC from Locality III; (b) idem for specimen A06412.SPC from Locality II.

paths trending towards the origin anchored line fits were applied to isolate characteristic remanent magnetizations (CRMs) (see Fig. 6). When curved trends of the vector end points were identified remagnetization circles (RMCs) were obtained (see Fig. 4). For the specimens that could not be demagnetized stable remanent magnetizations (SRMs) were calculated using Fisher's statistics (Fisher, 1953; see Fig. 3). End mean directions (EMDs) per site were obtained applying the combined analysis of remagnetization circles and stable remanent magnetizations of McFadden and McElhinny (1988), or Fisher's statistics. For the calculation of the EMDs only CRMs with maximum angular deviation (MAD) $\leq 15°$, SRMs with circles of 95% confidence (α_{95}) $\leq 10°$ and RMCs with $\lambda \leq 0.055$ were considered. Considering the low number of independently oriented samples per site, for the following analysis EMDs were rejected if their α_{95} exceeded 30° (Table 1).

A statistical test was used (McFadden and Lowes, 1981) to determine if the EMDs per site obtained for each locality have a common mean direction, i.e. if the EMDs calculated for Locality II and Locality III come from the same or from different populations. This test could not be performed for Locality I because only one EMD passed the selection criterion. At the 95% confidence level, the EMDs have the same mean direction. This means that they would come from the same population. Then, the EMDs obtained for each locality were analyzed together to calculate a single pole position.

The EMDs had normal and reverse polarities, so the reversal test (McFadden and McElhinny, 1990) was performed. With the in-situ EMDs the angle between the mean of the normal polarity set and the mean of the reverse polarity set was 3.8° and the critical angle was 9.8° ($N = 14$). With the tilt-corrected EMDs the angle between the mean of the normal polarity set and the mean of the reverse polarity set was 1.8° and the critical angle was 9.1° ($N = 14$), suggesting that the isolated magnetization would be pretectonic. The test was successful with classification B. This successful reversal test supports that there is no bias due to inappropriate analysis or deficient demagnetization and that overall the obtained EMDs fully average secular variations and reflect the axial dipole field.

The tilt test (McFadden, 1990) was also performed. The value of the precision parameter (K) was higher after structural correction. At the 95% confidence level the isolated magnetization could be considered as pretectonic (Table 2).

Then, the EMDs were converted to virtual geomagnetic poles (VGPs) and the mean pole position was calculated using Fisher's statistics:

Lat. = 77.4°S; Long. = 195.4°E;

$A_{95} = 4.7°$; $N = 14$; $K = 73$

The rotation represented by the palaeopole was obtained using as reference the South American reference pole determined by Roperch and Carlier

Table 1
Site mean directions of remanent magnetization, mean pole position and rotation determined for unit A

Site	N (No.)	Tilt corrected EMDs					Structure	
		decl.	incl.	α_{95}	K	polarity	strike	dip
Locality I								
6	3(3)	11.5°	−33.8°	21.9°	63	N	355°	15°
Locality II								
27	6(7)	191.9°	34.6°	20.0°	19	R	0°	0°
26	5(5)	188.7°	38.4°	7.6°	215	R	0°	0°
25	5(5)	5.4°	−39.9°	7.7°	100	N	0°	0°
24	1(3)	14.1°	−30.0°	–	–	N	310°	2°
23	3(3)	12.8°	−38.2°	5.7°	477	N	0°	0°
22	5(3)	194.3°	47.2°	16.7°	46	R	0°	0°
21	3(3)	26.0°	−48.1°	22.9°	30	N	290°	7°
Locality III								
18	2(2)	181.3°	34.5°	22.3°	85	R	360°	14°
16	3(3)	184.3°	39.7°	29.1°	44	R	340°	11°
15	2(2)	190.1°	43.6°	28.1°	54	R	340°	11°
13	3(3)	190.9°	34.5°	29.0°	22	R	25°	9°
11	1(1)	217.9°	40.5°	–	–	R	45°	8°
10	4(3)	202.0°	41.7°	15.4°	37	R	12°	9°

Mean pole position: Lat. 77.4°S, Long. 195.4°E, A_{95} 4.7°
Rotation: $R \pm \Delta R = 12.4 \pm 4.8°$

N (No.), number of specimens used for calculation (number of samples originally collected). EMDs (decl., incl., α_{95}, *K*, polarity) mean declination, mean inclination, radius of circle of 95% confidence, precision parameter and polarity of the end mean directions isolated per site. N and R are normal and reverse polarity, respectively. Mean pole position (lat., long., A_{95}) south latitude, east longitude and radius of circle of 95% confidence of the mean pole position. *R* and ΔR are rotation (clockwise positive) and confidence interval (Beck et al., 1986).

Table 2
Statistical values of the fold test of McFadden (1990)

	Declination	Inclination	*K*	α_{95}	SCOS2	*N*
In situ	197.3°	37.4°	74.8	4.6°	4.845	14
Tilt corrected	193.4°	39.2°	82.6	4.4°	1.305	14

Declination: declination of the end mean direction calculated for the sampled sections; Inclination: inclination of the end mean direction; *K*: presition parameter of the end mean direction; α95: confidence interval of the end mean direction; SCOS2: value of the statistical test (95 and 99% critical values are 4.358 and 6.087, respectively); *N* is number of sites considered.

(1992): Lat. = 80.5°S; Long. = 287.3°E; $A_{95} = 2.7°$ and applying the statistics of Beck et al. (1986):

$R \pm \Delta R = 12.4 \pm 4.8°$

4. Discussion

The obtained rotation value would indicate that the sampled zone underwent a clockwise vertical axis rotation of about 7–18°.

Localities II and III are within a distance of only 5 km and there is a major NNE–SSW-trending thrust fault to the west of the studied area. As it was mentioned above, this fault is the southern end of the eastern boundary of the granitic rocks of the Cumbres de Macón block (which underwent a tilt of 70° westward) and probably developed near the Miocene–Pliocene boundary. Thus the sampled rocks could record a local rotation induced by Miocene–Pliocene compression and accommodated by this fault zone.

On the other hand, two Neogene deformation phases are recognized in the southern Puna (Jordan and Alonso, 1987): an older deformation (Miocene–Pliocene), characterized by thrust faulting, produced

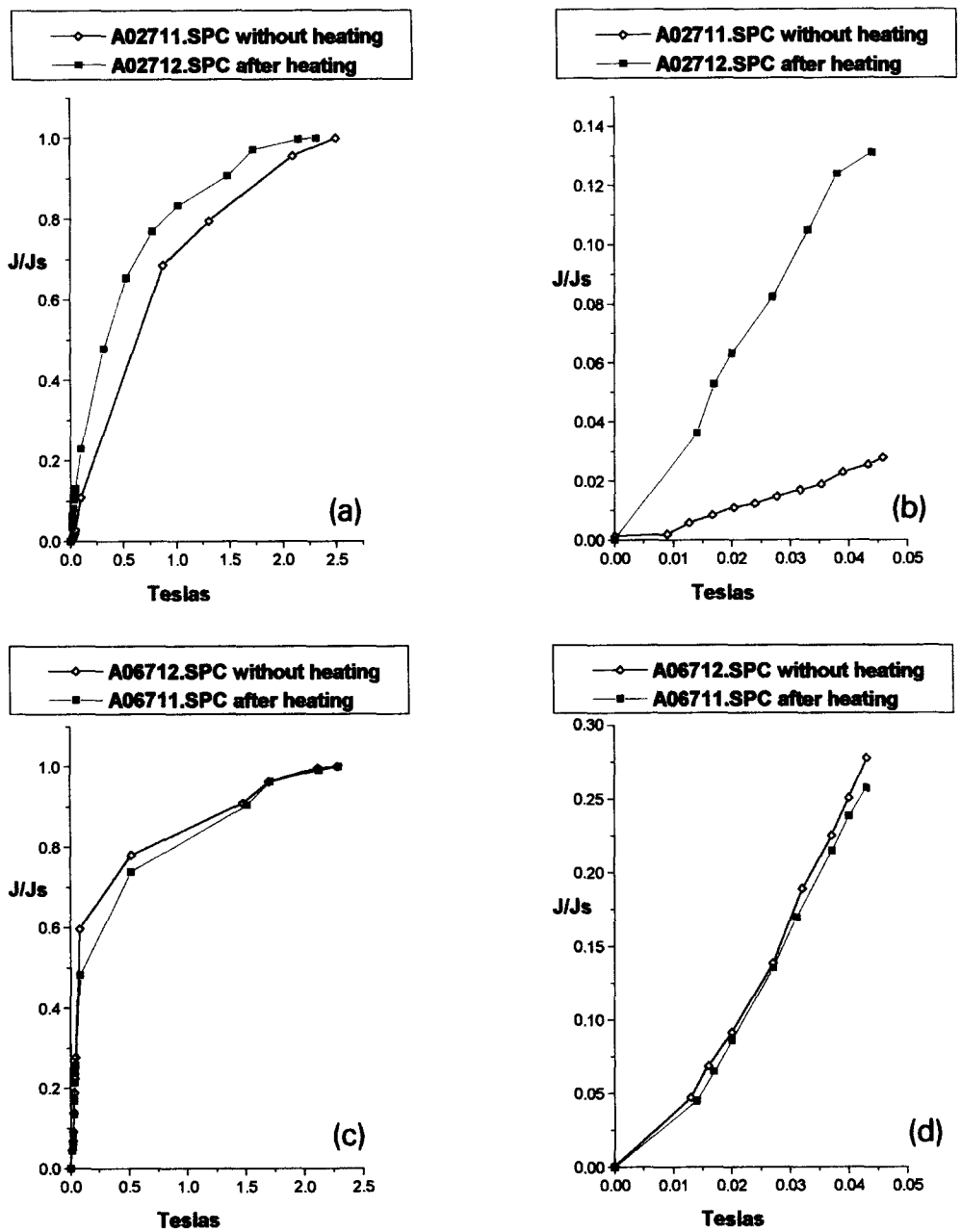


Fig. 9. IRM acquisition curves for specimen A02711.SPC (unheated) and A02712.SPC (after heating to 700°C) from Locality III (a, b), and idem for specimen A06712.SPC (unheated) and A06711.SPC (after heating to 680°C) from Locality II (c, d). (a and c) Results in high fields. (b and d) Results in low fields.

horizontal NW–SE shortening and vertical extension, and a younger deformation (Plio–Quaternary) that shows more diversity, but is characterized by NW–SE extension and SW–NE shortening. The younger shortening is parallel to the plate convergence direction during the last 10 Ma, but the older one is not, although both phases of deformation occurred during the same period of ENE plate con-

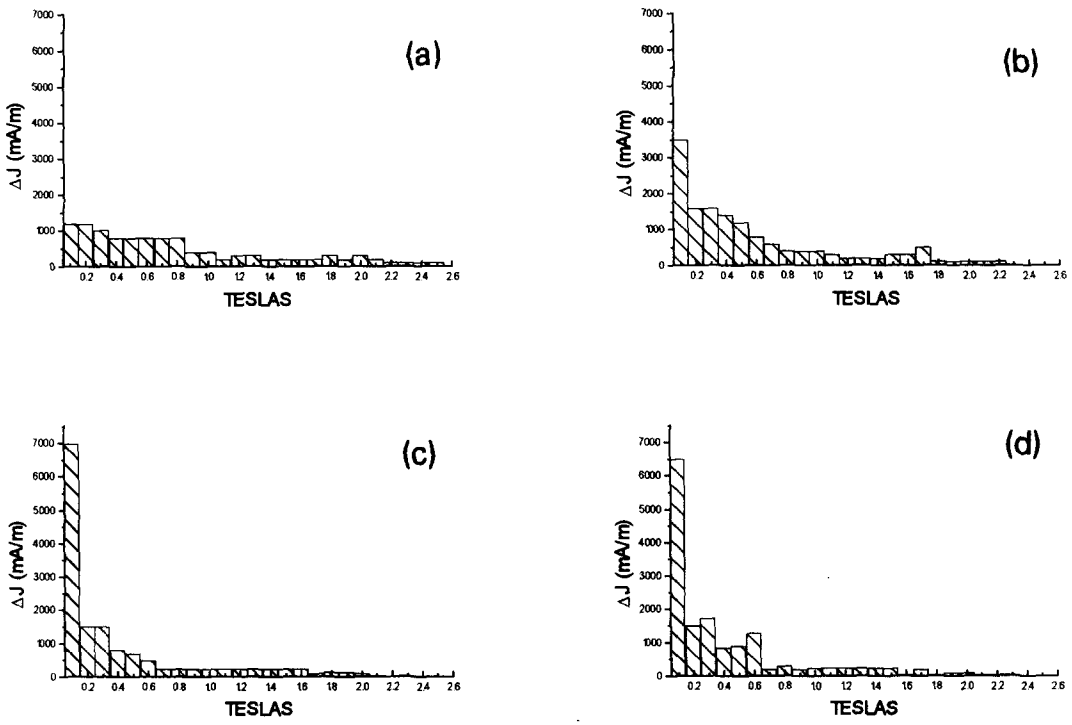


Fig. 10. (a) Coercivity spectrum (plotted as histogram) of IRM acquisition for specimen A02711.SPC (without heating). (b) Idem for specimen A02712.SPC (after heating). (c) Idem for specimen A06712.SPC (without heating). (d) Idem for specimen A06711.SPC (after heating).

vergence. In the studied area the Cumbres de Macón thrust fault (general strike NNE–SSW) developed during Miocene–Pliocene deformation. Clockwise vertical axis rotations of at least 40° are necessary to explain the actual orientation of the Miocene–Pliocene shortening (Marrett et al., 1994). The magnitude of the clockwise rotation determined in this study is less than the value mentioned above. Alternatively, the possibility that Miocene–Pliocene structures developed as reactivations of previous structural trends (Somoza et al., 1996b) could be considered. It is worthy to note that Miocene–Pliocene shortening is practically perpendicular to the Ordovician fold axes found in the Argentine Puna (Marrett et al., 1994) and that several authors have noticed the importance of ancient crustal weaknesses in the development of the Andean tectonics (e.g., Mon, 1979; Marrett et al., 1994). In that case, block rotations could be a response to distributed horizontal shear (Beck, 1988; Dewey and Lamb, 1992) resulting from the partition of the convergence vector between the Nazca and South America plates, since this vector was oblique to the continental margin during the Cenozoic (Pardo Casas and Molnar, 1987). This vector would have produced components perpendicular and parallel to previous structural features, where the tangential components would be right-lateral to the south and left-lateral to the north of Arica, explaining the sense of the observed rotations. By the way, it is very important to note that for a conclusive determination of the processes leading to the CARP a much more solid palaeomagnetic data base from the Central Andes is necessary along with detailed structural studies of the sampling areas.

5. Conclusions

A clockwise vertical axis rotation of about 7–18° has been determined for the Cenozoic sedimentary rocks which crop out in the southern Argentine Puna at about 24°30′S and 67°00′W.

Taking into account the presence of a major thrust

fault to the west of the studied area, this rotation could be of local character, induced by Miocene–Pliocene compression.

A much more solid palaeomagnetic database for the Central Andes is necessary together with a detailed structural control of the sampling areas in order to definitively determine the driving processes leading to the rotation pattern of the Central Andes.

Acknowledgements

We wish to thank especially Dr. Milka K. de Brodtkorb for her careful observation of polished sections and Dr. Rubén Somoza for his helpful suggestions. Utah Mining provided logistical support during the field work. This work was supported by the CONICET. The Universidad de Buenos Aires provided the necessary equipment. We thank P. Roperch and R.F. Burmester for helpful reviews of this paper.

References

Alonso, R.N., Jordan, T., Tabutt, K., Vandervoort, D., 1991. Giant evaporite belts of the Neogene Central Andes. Geology 19, 401–404.

Aubry, L., Roperch, P., Urreiztieta, M., Rossello, E., Chauvin, A., 1996. A paleomagnetic study along the south-eastern edge of the Altiplano–Puna Plateau: Neogene tectonic rotations. J. Geophys. Res. 101, 17883–17889.

Beck, M.E., 1988. Analysis of Late Jurassic–recent palaeomagnetic data from active margins of South America. J.S. Am. Earth Sci. 1, 39–52.

Beck, M.E., Burmester, R.F., Craig, D.E., 1986. Paleomagnetism of the Middle Tertiary volcanic rocks from the Western Cascade Series, northern California. J. Geophys. Res. 91, 8219–8230.

Beck, M.E., Burmester, R.F., Drake, R.E., Riley, P., 1994. A tale of two continents: some tectonic contrasts between the Central Andes and the North American Cordillera, as illustrated by their paleomagnetic signatures. Tectonics 13 (1), 215–224.

Butler, R.F., Marshall, L.G., Drake, R.E., Curtis, G.H., 1984. Magnetic polarity stratigraphy and ^{40}K–^{40}Ar dating of late Miocene and early Pliocene continental deposits, Catamarca province, NW Argentina. J. Geol. 92, 623–636.

Butler, R.F., Richards, D.R., Sempere, T., Marshall, L.G., 1995. Paleomagnetic determinations of vertical-axis tectonic rotations from Late Cretaceous and Paleocene strata of Bolivia. Geology 23 (9), 799–802.

Carey, S., 1955. The orocline concept in geotectonics. Proc. R. Soc. Tasmania 89, 255–288.

Dankers, P., 1978. Magnetic Properties of Dispersed Natural Iron-Oxides of Known Grain-Size. Ph.D. Thesis, Rijksuniversiteit, Utrecht, 143 pp. (unpubl.).

Dewey, J.F., Lamb, S.H., 1992. Active tectonics of the Andes. Tectonophysics 205, 79–95.

Dunlop, D., 1972. Magnetic mineralogy of unheated and heated red sediments by coercivity spectrum analysis. Geophys. J.R. Astron. Soc. 27, 37–55.

Fernández Garrasino, C., Bianucci, H., Musmarra, J., 1984. Algunos rasgos geológicos del sur de Salta y este de Tucumán. Bol. Inf. Pet. 1, 62–72.

Fisher, R., 1953. Dispersion on a sphere. Proc. R. Soc. London, Ser. A 217, 295–305.

Gubbels, T.L., Isacks, B.L., Farrar, E., 1993. High level surfaces, plateau uplift and foreland development, Bolivian Central Andes. Geology 21, 695–698.

Isacks, B.L., 1988. Uplift of the Central Andean Plateau and bending of the Bolivian Orocline. J. Geophys. Res. 93, 3211–3231.

Jordan, T., Alonso, R., 1987. Cenozoic stratigraphy and basin tectonics of the Andes Mountains 20–28° South Latitude. Am. Assoc. Pet. Geol. Bull. 71, 49–64.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geophys. J.R. Astron. Soc. 62, 699–718.

Koukharsky, M., 1988. Geologia de la Puna en la Región que media entre el Cerro Socompa y el Cerro Tul Tul, Provincia de Salta. Ph.D. Thesis, Universidad de Buenos Aires, 80 pp. (unpubl.).

Lowrie, W., 1990. Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties. Geophys. Res. Lett. 17 (2), 159–162.

MacFadden, B.J., Anaya, F., Perez, H., Naeser, C.W., Zeitler, P.K., Campbell Jr., K.E., 1990. Late Cenozoic paleomagnetism and chronology of Andean basins of Bolivia: evidence for possible oroclinal bending. J. Geol. 98, 541–555.

MacFadden, B.J., Anaya, F., Argollo, J., 1993. Magnetic polarity stratigraphy of Inchasi: a Pliocene mammal-bearing locality from the Bolivian Andes deposited just before the Great American Interchange. Earth Planet. Sci. Lett. 114, 229–241.

MacFadden, B.J., Anaya, F., Swisher II, C., 1995. Neogene paleomagnetism and oroclinal bending of the Central Andes of Bolivia. J. Geophys. Res. 100 (B5), 8153–8167.

Marrett, R., Allmendinger, R., Alonso, R., Drake, R., 1994. The Late Cenozoic tectonic evolution of the Puna Plateau and adjacent foreland, Northwestern Argentine Andes. J. S. Am. Earth Sci. 7 (2), 179–207.

McFadden, P.L., 1990. A new fold test for paleomagnetic studies. Geophys. J. Int. 103, 163–169.

McFadden, P.L., Lowes, F., 1981. The discrimination of mean directions drawn from Fisher distributions. Geophys. J.R. Astron. Soc. 67, 19–33.

McFadden, P.L., McElhinny, M.W., 1988. The combined analysis of remagnetization circles and direct observations in palaeomagnetism. Earth Planet. Sci. Lett. 87, 161–172.

McFadden, P.L., McElhinny, M.W., 1990. Classification of the reversal test in palaeomagnetism. Geophys. J. Int. 103, 725–729.

Méndez, V., 1974. Estructuras de las Provincias de Salta y Jujuy a partir del meridiano 65°30′ oeste, haste el límite con las repúblicas de Bolivia y Chile. Rev. Asoc. Geol. Arg. 29 (4), 391–424.

Mon, R., 1979. Esquema Tectónico de los Andes del Norte Argentino. Rev. Asoc. Geol. Arg. 34, 53–60.

Oviedo, E.S., 1989. MAG88: Un sistema de computación pare análisis de datos paleomagnéticos. Su aplicación al estudio paleomagnético de sedimentos cretácicos de la Cuenca Neuquina. Ph.D. Thesis, Universidad de Buenos Aires, 178 pp. (unpubl.).

Pardo Casas, F., Molnar, P., 1987. Relative motion of the Nazca Farallon) and South America plates since Late Cretaceous time. Tectonics 6, 233–248.

Randall, D., Taylor, G., Grocott, J., 1996. Major crustal rotations in the Andean margin: paleomagnetic results from the Coastal Cordillera of northern Chile. J. Geophys. Res. 101, 15783–15798.

Roperch, P., Carlier, G., 1992. Paleomagnetism of Mesozoic rocks from the central Andes of southern Perú: importance of rotations in the development of the Bolivian Orocline. J. Geophys. Res. 97, 17233–17249.

Roperch, P., Fornari, M., Herail, G., 1993. A paleomagnetic study of the Altiplano (extended abstract). Symp. Int. Géodyn. Andine, 2nd, pp. 241–244.

Schwarz, E.J., 1969. Thermomagnetic analysis of some red beds. Earth Planet. Sci. Lett. 5, 333–338.

Schwarz, E.J., 1970. Magnetochemical aspects of the heating of red and green beds. Geol. Surv. Can. Pap. 70-63, 31–34.

Singer, S., Puente, N., Sinito, A., Allmendinger, R., 1994. Estudio Paleomagnético de Secuencias Terciarias Aflorantes al Este del Salar de Antofalla, Puna Austral, Argentina. 7° Congr. Geol. Chileno, Concepción, Actas II, pp. 1451–1455.

Somoza, R., 1994. South American reference pole for the mid-Cretaceous: further constraints in the interpretation of Andean palaeomagnetic data. Geology 22, 933–936.

Somoza, R., Singer, S., Coira, B., 1996a. Paleomagnetism of upper Miocene ignimbrites at the Puna, an analysis of vertical-axis rotations in the Central Andes. J. Geophys. Res. 101, 11387–11400.

Somoza, R., Prezzi, C., Singer, S., 1996b. Análisis Tectónico de los Datos Paleomagnéticos del Terciario Superior de los Andes Centrales: Una Revisión del Oroclino Boliviano. XIII Congr. Geol. Argentino, III Congr. Exploración de Hidrocarburos, Buenos Aires, Actas II, pp. 493–504.

Stephenson, A., 1967. The effect of heat treatment on the magnetic properties of the Old Red Sandstone. Geophys. J.R. Astron. Soc. 13, 425–440.

Turner, P., 1980. Continental Red Beds. Developments in Sedimentology. Elsevier, Amsterdam, pp. 562.

Vandervoort, D., 1993. Non-Marine Evaporite Basin Studies, Southern Puna Plateau, Central Andes. Ph.D. Thesis, Cornell University (unpubl.).

Vandervoort, D., Jordan, T., Zeitler, P., Alonso, R., 1995. Chronology of internal drainage development and uplift, southern Puna Plateau, Argentine central Andes. Geology 23 (2), 145–148.

Watts, A.B., Lamb, S.H., Fairhead, J.D., Dewey, J.F., 1995. Lithospheric flexure and bending of the Central Andes. Earth Planet. Sci. Lett. 134, 9–21.

Zijderveld, J.A.A., 1967. AC demagnetization of rocks: analysis of results. In: Collinson, D.W., Creer, K.M., Runcorn, S.K. (Eds.), Methods in Paleomagnetism. Elsevier, Amsterdam, pp. 254–286.